

Axioms of Probability

1. Non-negativity: For any event A , $P(A) \geq 0$.
2. Normalization: $P(S) = 1$, where S is the sample space.
3. Additivity: For any two mutually exclusive events A and B , $P(A \cup B) = P(A) + P(B)$. This can be extended to $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ for mutually exclusive events $A_1, A_2, \dots, A_n$.

Basic Properties of Probability:

$$\forall A \in S, P(A) < 1$$

$$P(A^c) = 1 - P(A)$$

$$P(\emptyset) = 0$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$A \subseteq B \implies P(A) \leq P(B)$$

Counting:

Permutation — An arrangement of objects in a specific order. $P(n, r) = \frac{n!}{(n-r)!}$

Combination — A selection of objects without regard to the order. $C(n, r) = \frac{n!}{r!(n-r)!}$

Binomial Coefficient — The number of ways to choose k successes in n independent Bernoulli trials. $\binom{n}{k} = C(n, k) = \frac{n!}{k!(n-k)!}$

Number of Subsets — The number of subsets of a set with n elements is 2^n .

Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplication Rule: — $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$

$$P(\bigcap_{i=1}^n A_i) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_n|\bigcap_{i=1}^{n-1} A_i)$$

Law of Total Probability: — If $B_1, B_2, \dots, B_n$ are mutually exclusive and exhaustive events, then for any event A : $P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$

Bayes' Theorem: —

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$$

Independence:

Two events A and B are independent if and only if $P(A|B) = P(A)$ and $P(A \cap B) = P(A)P(B)$.

If A and B are independent, then B and A , A and B^c , A^c and B , A^c and B^c are also independent.

Random Variables An RV is a function that assigns a real number to each outcome in the sample space of a random experiment.

Discrete RV — A random variable that can take on a countable number of distinct values.

Continuous RV — A random variable that can take on any value within a given range or interval.

$X : S \rightarrow \mathbb{R}$ Where S is the event space and X is the random variable.

PMF — Probability Mass Function for discrete RVs. $P(X = x)$

CDF — Cumulative Distribution Function for both discrete and continuous RVs. $F(x) = P(X \leq x) = \sum_{i \leq x} p(i)$ This is

just like the pmf but cumulative rather than pointwise.

Bernoulli Random Variable — A discrete RV that takes the value 1 with probability p and the value 0 with probability $1 - p$.

Expected Value — The mean. $E(X) = \mu_X = \sum_{-\infty}^{\infty} xP(X = x)$ for discrete RVs and $E(X) = \mu_X = \int_{-\infty}^{\infty} xf(x)dx$ for continuous RVs.

$E(h(X)) = \sum_x h(x)P(X = x)$ for discrete RVs and $E(h(X)) = \int_{-\infty}^{\infty} h(x)f(x)dx$ for continuous RVs.

$$E(aX + b) = aE(X) + b$$

$$E(X + Y) = E(X) + E(Y)$$

$$E(X^2) = \sum_x x^2 p(x)$$

Variance — A measure of how much the values of a random variable deviate from the mean. $Var(X) = \sigma_X^2 = E[(X - \mu_X)^2] = E(X^2) - (E(X))^2$

Standard Deviation — The square root of the variance. $\sigma_X = \sqrt{Var(X)}$

Common Discrete Distributions:

Binomial Distribution — Models the number of successes in n independent Bernoulli trials, each with success probability p .

$$X \sim Bin(n, p)$$

PMF: — $P(X = k) = b(x; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$ for $k = 0, 1, \dots, n$ else 0

$$\text{CDF:} \quad B(x; n, p) = P(X \leq x) = \sum_{y=0}^{\lfloor x \rfloor} b(y; n, p)$$

$$\text{Mean:} \quad E(X) = np$$

$$\text{Variance:} \quad Var(X) = np(1 - p)$$

Negative Binomial Distribution — Models the number of trials needed to achieve r successes in a sequence of independent Bernoulli trials, each with success probability p .

$$X \sim NB(r, p)$$

PMF: — $P(X = k) = \binom{k-1}{r-1} p^r (1 - p)^{k-r}$ for $k = r, r+1, \dots$ else 0

$$\text{CDF:} \quad NB(x; r, p) = P(X \leq x) = \sum_{y=r}^{\lfloor x \rfloor} \binom{y-1}{r-1} p^r (1 - p)^{y-r}$$

$$\text{Mean:} \quad E(X) = \frac{r}{p}$$

$$\text{Variance:} \quad Var(X) = \frac{r(1-p)}{p^2}$$

Geometric Distribution — A special case of the negative binomial distribution where $r = 1$. It models the number of trials needed to get the first success.

$$X \sim Geo(p)$$

PMF: — $P(X = k) = (1 - p)^{k-1} p$ for $k = 1, 2, \dots$ else 0

$$\text{CDF:} \quad G(x; p) = P(X \leq x) = 1 - (1 - p)^{\lfloor x \rfloor}$$

$$\text{Mean:} \quad E(X) = \frac{1}{p}$$

$$\text{Variance:} \quad Var(X) = \frac{1-p}{p^2}$$

Hypergeometric Distribution — Models the number of successes in n draws from a finite population of size N containing K successes, without replacement.

$$X \sim Hyp(N, K, n)$$

PMF: — $h(x; n, N, K) = P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$ for $\max(0, n - (N - K)) \leq k \leq \min(n, K)$ else 0

$$\text{CDF:} \quad H(x; n, N, K) = P(X \leq x) =$$

$$\sum_{y=\max(0, n-(N-K))}^{\lfloor x \rfloor} \frac{\binom{K}{y} \binom{N-K}{n-y}}{\binom{N}{n}}$$

Mean: — $E(X) = n \frac{K}{N}$

Variance: — $Var(X) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$

Poisson Distribution — Models the number of events occurring in a fixed interval of time or space, given the average rate λ of occurrence.

Used to model the number of times a rare event occurs in a large population.

$$X \sim Pois(\lambda)$$

PMF: — $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \dots$ else 0

CDF: — $P(X \leq x) = \sum_{y=0}^{\lfloor x \rfloor} \frac{\lambda^y e^{-\lambda}}{y!}$

Mean: — $E(X) = \lambda$

Variance: — $Var(X) = \lambda$